

Principles and Tools in Biological Data Analytics I

Exercise Sheet 1 — 18 September 2026

Problem 1

Write code to print the first 10 natural numbers to the console using a while loop.

Problem 2

Write code to print the following pattern to the console using the loop concept.

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

Problem 3

Write code to calculate the sum of all numbers from 1 up to a given number n .

Problem 4

Write code to print the formatted multiplication table for a given number n .

Problem 5

Write code to display only specific items from a list/array that satisfy the following conditions:

- The number must be divisible by five.
- If the number is greater than or equal to 150, then skip it.
- If the number is greater than 500, then stop the loop.

Example input:

- Python: [12, 75, 150, 182, 145, 525, 50]

Problem 6

Write code to count the digits of a given number n . The code should print the result to the console. For example, for $n = 75869$ the output should be 5.

Problem 7

Write code to print all prime numbers between any given pair of numbers n and m to the console. Use as example input $n = 25$ and $m = 50$.

NOTE: A prime number is a natural number greater than 1 that can only be divided by itself and 1 without remainders. Examples are: 2, 3, 5, 7, 11.

Problem 8

Write code that reverses a given string number s . For example “stressed“ should result in “desserts“. Similarly, write code that reverses a given integer n . For example $n = 76542$ should result in $m = 24567$.